Temporal LP economics — answers to your questions

Scope: the **protected perps AMM** (not the CLOB-AMM). Every exhibit below is **live** in `temporal_lp_economics_MODEL_v5_options_seller.xlsx` (cell ref given per exhibit — edit the yellow **Assumptions** and all exhibits recompute). Brainstorm; magnitudes illustrative.

Q1. How does the BASE AMM LP perform across regimes?

> **Short-volatility** position: the LP sells the perpetual-option book (bands), delta-hedged on Hyperliquid. It **profits when realized vol comes in below the calibration vol**, loses in spikes.

Exhibit A — BASE net LP APR (rows = realized vol, cols = calibration vol · green = profit / red = loss)

RV \ σ_cal	30%	50%	70%	90%	110%
20%	16%	64%	136%	232%	352%
40%	-20%	28%	100%	196%	316%
60%	-80%	-32%	40%	136%	256%
80%	-164%	-116%	-44%	52%	172%
100%	-272%	-224%	-152%	-56%	64%

`↳ live in spreadsheet:` **`Regime_Grid!B5:F9 (axes A5:A9, B4:F4)`**

Q2. How does the RESTAKED (HLP-margined) LP perform?

> Same shape **+ a flat +10%/yr HLP yield** (additive, un-levered) → lifts every cell, **widening the profitable region**.

Exhibit B — RESTAKED net LP APR (= Base + HLP yield)

RV \ σ_cal	30%	50%	70%	90%	110%
20%	26%	74%	146%	242%	362%
40%	-10%	38%	110%	206%	326%
60%	-70%	-22%	50%	146%	266%
80%	-154%	-106%	-34%	62%	182%
100%	-262%	-214%	-142%	-46%	74%

`↳ live in spreadsheet:` **`Regime_Grid!B14:F18`**

Q3. AMMs are grossly lossmaking — how does THIS one make money?

> It is **not a passive spot LP** (which bleeds LVR). It is an **option seller**: collects the **funding-carry** (priced at the calibration vol), pays the realized-vol gamma cost, delta-hedged. Edge = **selling vol above what realizes** + fees + HLP yield. Loses only in vol spikes / mis-calibration. The money is **carry − realized vol cost**, not passive fees.

Exhibit C — P&L build-up (scenario: realized vol 60%, calibration vol 70%; per $ equity, annualized)

Carry income (funding, at σ_cal)	147.0%	B7
− Vol cost (realized gamma bleed = LVR-equiv)	-108.0%	B8
= Options net	**39.0%**	B9
+ Fee income	2.7%	B10
− Hedge/exec/order costs	-1.5%	B12
= BASE LP net	**40.2%**	B13
+ HLP yield (restaked)	10.0%	B14
= RESTAKED LP net	**50.2%**	B15

`↳ live in spreadsheet:` **`Scenario_calc!A6:B15 (inputs B3:B4)`**

Caveats: reduced-form sketch, not a backtest; carry magnitude rests on the real funding law (TBD) and the gamma coefficient needs calibration; well-calibrated with no vol edge the LP nets only ~+1%/yr (fees − costs) — the return **requires calibrating vol above realized**. Assumptions are on the `Assumptions` sheet. Brainstorm / non-core.